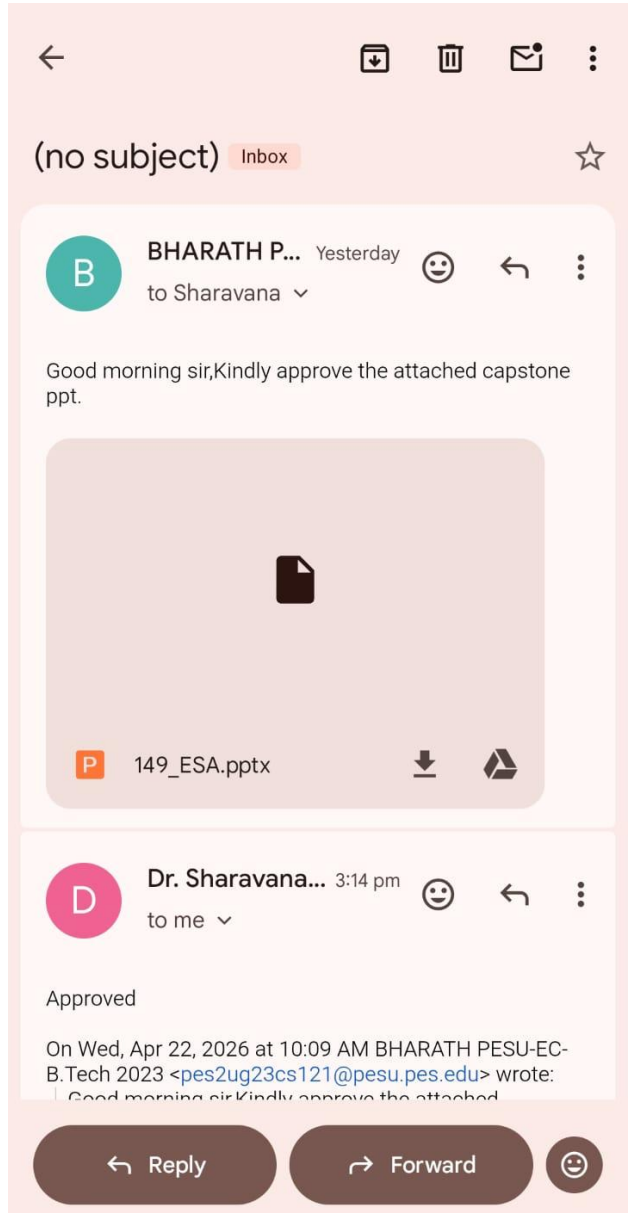

(no subject)
Inbox
B
BHARATH P...
Yesterday
to Sharavana
Good morning sir,Kindly approve the attached capstone ppt.
149_ESA.pptx
D
Dr. Sharavana...
3:14 pm
to me
Approved
On Wed, Apr 22, 2026 at 10:09 AM BHARATH PESU-EC-B.Tech 2023 <pes2ug23cs121@pesu.pes.edu> wrote:
Reply
Forward

UE23CS320B – ESA REVIEW

Project Title : **Intelligent Energy Management System (IEMS)**

Project ID : 149

Project Guide : Dr. Sharavana K

Project Team : BASAVA DARSHAN M | PES2UG23CS118

BHANUTEJA MANCHAL | PES2UG23CS119

BHARATH | PES2UG23CS121

BHARGAV S | PES2UG23CS124

Problem Statement and Abstract

Problem Statement

- Most homeowners only have access to monthly electricity bills, whereas accurate solar and battery system design requires detailed hourly consumption data. This data gap often results in improper system sizing, inefficient battery usage, and unreliable financial estimates.

- Additionally, optimizing battery operation under varying weather conditions is complex. Hence, there is a need for an intelligent simulation framework that can generate realistic load data, optimize system configuration, and evaluate techno-economic performance effectively.

Project Abstract

The Intelligent Energy Management System (IEMS) is an AI-driven framework designed to resolve the pervasive data deficit in residential renewable energy planning.
By leveraging probabilistic modeling techniques, IEMS synthesizes realistic hourly load profiles from minimal data. The system utilizes evolutionary algorithms to optimize hardware capacity under financial constraints.
Finally, the framework employs Deep learning within a deterministic, physics-based environment to simulate intelligent battery dispatch, ensuring highly accurate techno-economic analysis.

Suggestions from Review-1 and Review-2

Following the preliminary review, the following adjustments have been integrated into the project and presentation .

1. Reframing the Project Title

Panel Feedback: The previous title was deemed overly generic.

2. Enhancing Presentation

Panel Feedback: Slide materials lacked professional and academic depth.

Analysis and Critique of Research/Relevance - Application in Real world

Relevance & Application in Real World

The IEMS framework moves beyond rudimentary "Rule of Thumb" methodologies into dynamic AI simulation:

- **Solar Installation Contractors (B2B SaaS):** Functions as a point-of-sale validation tool. Contractors utilize the Genetic Algorithm to state: *"Our AI tested 1,000 combinations to find the maximum ROI for your budget."*
- **Homeowner Validation:** Empowers the end-consumer with an objective analysis tool to audit quotes, preventing predatory upselling of excessive battery capacities.
- **Smart Grid Edge Deployment:** The DQN Learning logic can be flashed onto edge devices to actively control physical inverters based on real-time NASA weather APIs.

Analysis & Critique of Research

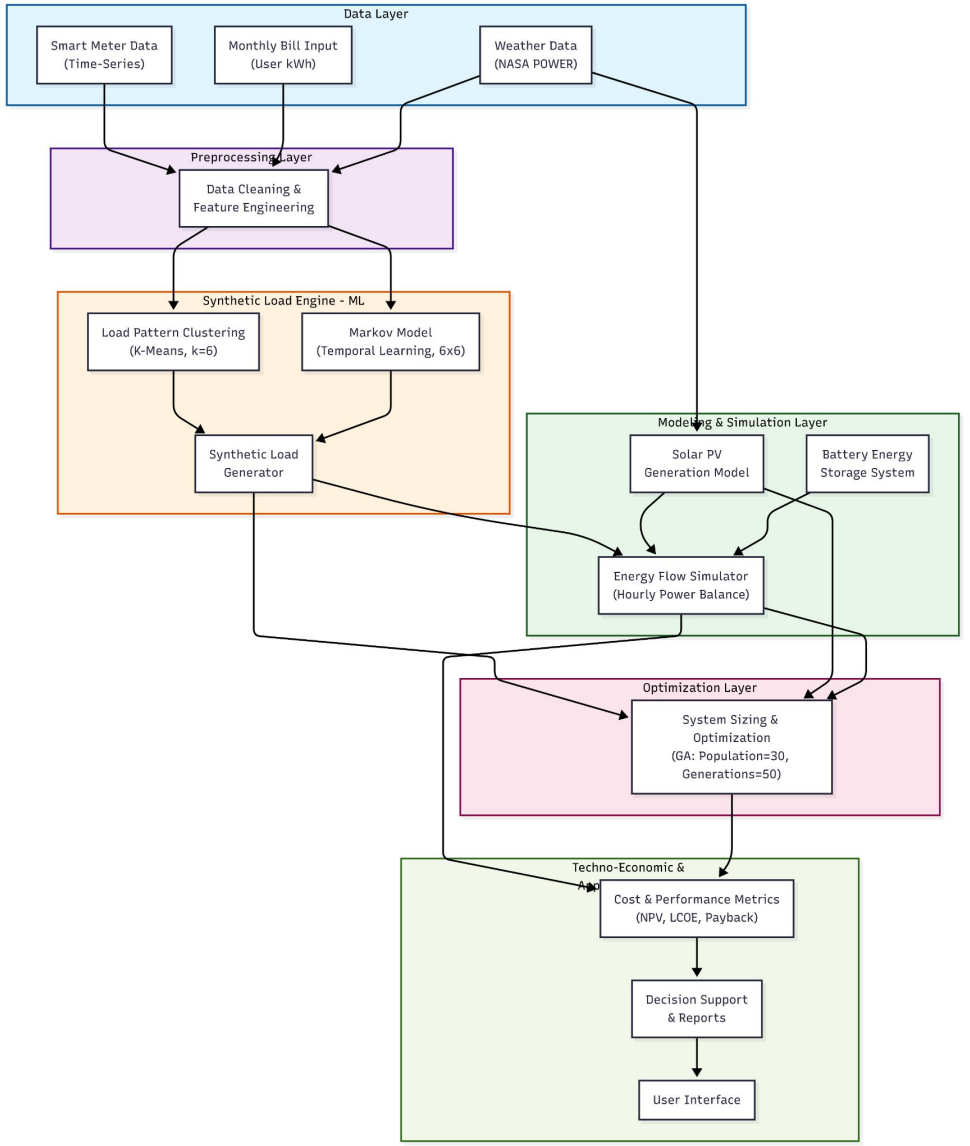
A rigorous analysis reveals three primary methodological constraints:

- **1. The Markov Assumption :** Synthetic generation relies on historical probabilities. If a structural shift occurs, the historical transition matrix fails to predict the new paradigm.
- **2. Micro-climate Constraints:** The NASA POWER API provides highly accurate regional irradiance but cannot account for localized, physical shading events (e.g., adjacent foliage or architecture).
- **3. AI Aggression vs. Degradation:** Deep Q-Networks inherently pursue maximum reward. Without severe penalties, the agent may aggressively cycle the battery, rapidly accelerating chemical degradation.

High Level Design

The system is architected as a decoupled pipeline, cleanly separating data synthesis, physical constraints, and control logic into modular units.

- **Load Synthesis :** Unsupervised learning pipeline ingesting minimal billing data to extrapolate probabilistic 720-hour matrices.
- **Physics Engine :** Deterministic core utilizing NASA API variables to bound system outputs within strict thermodynamic limits.
- **Optimization :** Sequential execution of Genetic Algorithms (static hardware sizing) followed by Deep Q-Networks (dynamic temporal dispatch).



Design Methodology

A. Hybrid Intelligent Methodology

The architecture actively avoids monolithic Deep Learning. By utilizing distinct models (K-Means/Markov for generation, GA for hardware search, DQN for control), the system ensures high modularity, computational efficiency, and mathematical interpretability.

B. Physics-Constrained Optimization

To eliminate infeasible configurations, optimization components are decoupled from yield calculation. The system must query the deterministic Physics Engine for all performance evaluations, ensuring that every configuration adheres to thermodynamic constraints.

C. Simulation-First SDLC

Development followed a bottom-up methodology. The digital twin (Physics Engine environment) was constructed prior to optimizer development. This ensured all algorithms were benchmarked against a functionally accurate physical baseline.

Project Progress

Current Status: 40% Completed (Data Synthesis & Physical Environment Locked)

The first two major milestones are fully operational. Development has now shifted to evolutionary hardware optimization (Brain 1b).

COMPLETED

1. Load Gen

Markov Chain logic deployed. Generates realistic 720-hour time-series data.

COMPLETED

2. Physics Engine

NASA API integrated. Calculates PV generation with strict physical constraints

IN PROGRESS

3. Sizing GA

Training evolutionary algorithm to find optimal hardware bounds via fitness score.

PENDING

4. DQN Controller

Training Deep Q-Network for intelligent hour-by-hour operational dispatch.

PENDING

5. UI & Validation

React.js visualization dashboard and final end-to-end benchmarking.

GA - Implementation

Implementation of the Genetic Algorithm (GA) Module for best solar and battery Sizing

1. Modular Architecture of the Optimization Engine

The Genetic Algorithm framework has been developed as a highly decoupled, modular package (`/optimization`). It programmatically discovers the optimal solar/battery hardware combination constrained strictly by user budgets.

- **Component Loader:** Ingests realistic datasets (15+ models of Solar Panels and Li-Ion batteries) containing cost, efficiency, and degradation metrics.
- **Design Space Generator:** Pre-calculates valid hardware boundaries. E.g., prevents testing a 20kWh battery if the user's budget is only ₹2,000.
- **CLI Interface:** An interactive terminal entry point (`main.py`) for users to define usage, budget, and geolocation cleanly.

System Directory Structure

```
optimization/
├── component_loader.py      # Parses CSV datasets
├── design_space.py          # Generates valid combinations
│
├── objective_functions.py # Fitness eval via simulator
├── optimizer.py             # Core GA algorithmic loop
├── main.py                  # CLI / User Entry Point
└
```

2. Algorithmic Pipeline: Process Flow

The optimization process executes a biologically-inspired evolutionary loop. It integrates directly with the physical Simulator to evaluate hardware viability over a synthetic time-series baseline.

1. Initialization

User inputs Budget (₹) & kWh. System loads component DB and generates a random, budget-compliant population.

2. Evolutionary Loop (N Gens)

- **Evaluate:** Run simulator to get Fitness.
- **Select:** Tournament selection of best.
- **Crossover:** Blend solar/battery genes.
- **Mutate:** small random changes in solar or battery size

Elitism: Preserve top 10% strictly.

3. Convergence

Outputs optimal kW and kWh capacities, maximizing ROI and outputs optimal solar and battery size along with savings

3. The Objective (Fitness) Function

The core intelligence of the GA relies on the Objective Function.
It penalizes grid reliance and rewards solar utilization, subject to a mathematical constraint: **System Cost ≤ User Budget**.

$$\text{Fitness} = (W_1 \times \text{Solar Utilization}) + (W_2 \times \text{Grid Reduction}) + \text{SoC Bonus}$$

Simulation Integration Parameters

- **Weights Configuration:** Standardized at 60% Solar Weight (W_1) and 40% Grid Independence Weight (W_2).
- **Dynamic Constraints:** If a mutation creates an individual exceeding the budget, it is immediately assigned a fitness of 0.
- **Battery Lifecycle Logic:** The function auto-calculates charge/discharge limits and provides an "SoC Bonus" for batteries operating in healthy thermodynamic ranges.

4. Practical Application & Outputs

The system is actively functional. Below demonstrates how the module adapts hardware recommendations across varying real-world user profiles and exports data.

Use Case: Residential

Input: 300 kWh/mo | Budget: ₹4L

Optimal Configuration:

- 6 kW Solar Array
- 10 kWh Battery Storage
- Result: Grid drop to <30%

Use Case: Commercial

Input: 800 kWh/mo | Budget: ₹12L

Optimal Configuration:

- 12 kW Solar Array
- 15 kWh Battery Storage
- Result: ROI achieved in 5 Yrs

Output Generation (JSON)

Results are saved to `latest_optimization_result.json` containing:

- Exact component IDs (e.g., SP_IND_010)
- Precise hardware counts
- Total estimated monetary savings
- Algorithm convergence statistics

Data Pipeline & Component Architecture

LEAD DATA ARCHITECT

- **Component Dataclass Engineering** Developed `component_loader.py` to ingest raw hardware CSV datasets (15+ market models) into strict Python objects, ensuring correct data types for efficiencies, degradation rates, and physical limits.
- **Hardware Quantization Logic** Constructed the mathematical foundation that allows the continuous outputs of the GA to be mapped to discrete, physical hardware items (e.g., converting a 5.2kW optimization output into exactly 13 physical 400W panels).
- **Temporal Data Synchronization** Managed the NASA POWER API extraction, handling complex timezone transformations (UTC to IST) to perfectly align physical solar irradiance profiles with local synthetic load data.

CORE CONCEPT: HARDWARE QUANTIZATION

To translate the optimization algorithm continuous capacity guesses into physical reality, raw data is quantized into discrete hardware units using a strict ceiling function:

$$N_{units} = \lceil \frac{C_{target}}{C_{spec}} \rceil$$

N_units: Discrete number of physical panels or batteries
C_target: Continuous capacity requested by the GA
C_spec: Rated capacity of the specific loaded hardware

Core Algorithmic Mathematics

ALGORITHM DEVELOPER

- **Evolutionary Logic Formulation** Programmed the overarching Genetic Algorithm (`optimizer.py`). Defined the Population dynamics (N=30) and the specific Generation lifecycle (50 epochs) necessary for mathematical convergence.
- **Continuous Blended Crossover** Rather than strictly swapping solar panels and batteries between parent solutions, implemented a continuous blending technique using an alpha coefficient to discover highly specific intermediate hardware capacities.
- **Elitism & Mutation Governance** Programmed the strict Elitism rule (preserving the top 10% fitness individuals) while maintaining a 10% mutation injection rate to prevent the algorithm from collapsing into local minima.

CORE CONCEPT: BLENDED GENE CROSSOVER

To explore the hardware design space more granularly, offspring capacities are calculated via an alpha-blend of parent parameters:

$$C_{kW} = \alpha \ P_{1,kW} + (1 - \alpha) \ P_{2,kW}$$

`C_kW`: Hardware capacity of the Child solution
`P_1, P_2`: Capacities of the Parent solutions
`α`: Random blending coefficient [0, 1]

Business Logic & Constraints

VALIDATION & CONSTRAINT ENGINEER

- **Design Space Pruning** Engineered `design_space.py` to calculate raw component costs against the user's maximum budget. Invalid hardware vectors are mathematically culled prior to initializing the Genetic Algorithm to save compute cycles.
- **The Objective (Fitness) Function** Authored the primary scoring formula. The system fundamentally balances two competing goals: maximizing the consumption of generated solar energy (60% weight) while aggressively minimizing reliance on the utility grid (40% weight).
- **Thermodynamic Battery Penalties** Integrated a State-of-Charge (SoC) bonus logic. Solutions that excessively discharge the battery below safe chemical limits lose their fitness bonus, protecting physical asset lifespan.

CORE CONCEPT: THE FITNESS EQUATION

The Genetic Algorithm seeks to maximize this overarching objective function, subject to the hard budget constraint:

$$F = W_s \times U + W_g \times R + \beta$$

F: Total Fitness Score (Max 100)

U: Solar Utilization % | **R**: Grid Reduction %

W_s, W_g: Tuned weights (0.6 and 0.4)

β: Battery Health/SoC Bonus

Backend Architecture & Integration

BACKEND INTEGRATOR

- **System Orchestration** main program that manages the full workflow. It connects the optimization algorithm with the system simulator, so every candidate solution is evaluated using a realistic, physics-based model. This makes the optimization results accurate and reliable.
- **Financial Translation Engine** Built the calculation framework to convert raw simulated kWh flows into exact Indian Rupee (₹) savings, heavily utilizing non-linear, tiered slab tariffs to reflect real-world consumer billing.
- **Artifact Generation (JSON)** system that saves the final best results in a clean JSON format. It stores the selected configuration and its performance details in a structured way. This makes it easy to use later, especially for a React.js frontend to display the data.

CORE CONCEPT: TIERED TARIFF PRICING

To accurately simulate financial Return on Investment (ROI), the bill calculation utilizes a piecewise summation across Indian tariff slabs:

$$\text{Cost} = \sum_{k=1}^{S} \min\left(E_{\text{rem}}, L_k\right) \times R_k$$

k: Individual Tariff Slab (e.g., 0-50 kWh)

E_rem: Remaining Energy to be billed

L_k: Max limit of the current slab

R_k: Cost Rate (₹) of the current slab

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Output Screenshot

```
(.venv) PS E:\Capstone Research papers\Capstone code files\Capstone_Project> python -m optimization.main

          INTELLIGENT ENERGY MANAGEMENT SYSTEM (IEMS)
                Genetic Algorithm Optimization Engine

  Optimize Solar + Battery Sizing for Maximum Savings & Grid Independence


================================================================================
  USER INPUTS
================================================================================

📋 Please provide the following information:

   Monthly electricity usage (kWh) [e.g., 300]: 300

   💡 Calculated Monthly Bill (Tiered Pricing): ₹1,775.00

   Total budget for solar system (₹) [e.g., 500000]: 500000

   Location [bangalore, mumbai, delhi...]: bangalore

   Number of days for load profile [default: 30]:

   Advanced Options (press Enter for defaults):
   Population size [default: 30]:
   Generations [default: 50]:
   Mutation rate [default: 0.1]:

================================================================================
  INITIALIZING OPTIMIZATION
================================================================================


================================================================================
  DATA GENERATION & PREPARATION
================================================================================


   💰 Monthly Consumption & Cost:
      Monthly Usage: 300.00 kWh
      Monthly Bill (Tiered): ₹1,775.00
      Analysis Period: 30 days
```

```
(.venv) PS E:\Capstone Research papers\Capstone code files\Capstone_Project> python -m optimization.main
======================================================================
OPTIMIZATION RESULTS
======================================================================

🎯 OPTIMAL SYSTEM CONFIGURATION
----------------------------------------------------------------------
  Solar Capacity: 9.3 kW (23 panels)
  Battery Capacity: 2.0 kWh (1 units)
  Total System Cost: ₹     499,500
  Budget Available: ₹     500,000
  Budget Utilization:        99.9%

📊 PERFORMANCE METRICS
----------------------------------------------------------------------
  Grid Dependency:                          55.0%
  Solar Utilization:                        32.6%
  Avg. Battery SoC:                         41.4%
  Estimated Savings: ₹                     1,319

🏆 FITNESS SCORE: 42.55/100
----------------------------------------------------------------------

📈 OPTIMIZATION STATISTICS
----------------------------------------------------------------------
  Generations: 50
  Population Size: 30
  Final Fitness: 42.55
  Initial Fitness: 42.22
  Improvement: +0.33

======================================================================


================================================================================
  RESULTS EXPORT
================================================================================

✓ Results saved to: E:\Capstone Research papers\Capstone code files\Capstone_Project\results\optimization_result_20260422_094028.json
✓ Latest result saved to: E:\Capstone Research papers\Capstone code files\Capstone_Project\results\latest_optimization_result.json

================================================================================
  FINAL SUMMARY
================================================================================

  📊 SYSTEM SIZING
  ------------------------------------------------------------------------------
    Solar System Size:        9.3 kW
```

Parameter	Budget 500000	Budget 1000000	Difference (1000000 - 500000)
Monthly usage (kWh)	300.0	300.0	0.0
Monthly bill (₹)	1,775.00	1,775.00	0.00
Valid candidates within budget	30	92	+62
Solar capacity (kW)	9.3	14.0	+4.7
Solar panels count	23	35	+12
Battery capacity (kWh)	2.0	7.0	+5.0
Battery units	1	1	0
Total system cost (₹)	499,500	980,000	+480,500
Budget allocated (₹)	500,000	1,000,000	+500,000
Budget remaining (₹)	500	20,000	+19,500
Budget utilization (%)	99.9%	98.0%	-1.9 pp
Grid dependency (%)	55.0%	25.0%	-30.0 pp
Solar utilization (%)	32.6%	35.9%	+3.3 pp
Avg battery SoC (%)	41.4%	50.9%	+9.5 pp
Estimated savings per month (₹)	1,319	601	-718
Annual savings (₹)	15,833.28	7,211.52	-8,621.76
Fitness score	42.55	56.55	+14.00
Generations	50	50	0
Population size	30	30	0
Initial fitness	42.22	53.02	+10.80
Final fitness	42.55	56.55	+14.00
Fitness improvement	0.33	3.52	+3.19

Thank You

Intelligent Energy Management System (IEMS) • Capstone Project

Individual Contribution

Team Member	Domain Focus	Specific Engineering Contributions
Member 1	Data Pipeline & Pre-processing	Engineered smart meter cleaning pipeline; managed NASA API extraction/normalization; designed Genetic Algorithm Chromosome data structures.
Member 2	Core Algorithmic Mathematics	Implemented K-Means clustering and Markov transition matrices; translated thermodynamic formulas; programmed GA Crossover/Mutation logic.
Member 3	Business Logic & Constraint Validation	Developed scaling logic for kWh matching; validated PV outputs against benchmarks; engineered strict ROI Fitness Function penalty parameters.
Member 4	Backend Architecture & Integration	Wrapped ML scripts into Object-Oriented classes; built asynchronous API hooks for weather routing; designed orchestration loop for GA evaluation.

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